



Schisandra chinensis (Turcz.) Baill. Protects against DSS-induced colitis in mice: Involvement of TLR4/NF- κ B/NLRP3 inflammasome pathway and gut microbiota

Zhenhua Bian^{a,b,1}, Yuwen Qin^{a,1}, Lin Li^{a,1}, Lianlin Su^a, Chenghao Fei^a, Yu Li^a, Minmin Hu^b, Xiaowei Chen^b, Wei Zhang^a, Chunqin Mao^a, Xiaohang Yuan^{b,*}, Tulin Lu^{a,*}, De Ji^{a,***}

^a College of Pharmacy, Nanjing University of Chinese Medicine, Nanjing, 210023, China

^b Department of Pharmacy, Wuxi TCM Hospital Affiliated to Nanjing University of Chinese Medicine, Wuxi, 214071, China

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ABSTRACT

Ethnopharmacological relevance: the fruit of *Schisandra chinensis* (Turcz.) Baill. (SC) is an important traditional Chinese herbal medicine, which has been widely used in traditional Chinese medicine (TCM) for treating intestinal diseases. It is also traditionally used as health product and medicine in Russia and other countries. However, the effect of SC ethanol extract on anti-ulcerative colitis (UC) has not been systematically studied yet. **Aim of the study:** We investigated the protective effects and underlying action mechanisms of SC extract (SCE) for UC treatment.

Materials and methods: An animal model of UC induced by dextran sulfate sodium (DSS) was established. After oral administration of SCE, the Disease Activity Index (DAI) was calculated, the length of colon measured, levels of proinflammatory factors determined, and histopathology carried out to assess the therapeutic efficacy of SCE on UC. The effects of SCE on the toll-like receptor 4/nuclear factor-kappa B/nucleotide-binding and oligomerization domain-like receptor family pyrin domain containing 3 inflammasome (TLR4/NF- κ B/NLRP3 inflammasome) signaling pathway were evaluated by western blotting. High-throughput sequencing was done to reveal the effect of SCE on the change of the gut microbiota (GM) in mice with DSS-induced colitis.

Results: SCE significantly reduced the DAI score, restored colon-length shortening, and ameliorated colonic histopathologic injury in mice with DSS-induced colitis. SCE inhibited the inflammatory response by regulating the TLR4/NF- κ B/NLRP3 inflammasome pathway in mice with UC. SCE also maintained gut barrier function by increasing the levels of zonula occludens (ZO)-1 and occludin. 16S rRNA sequencing showed that SCE could reverse the GM imbalance caused by UC.

Conclusions: SCE can ameliorate DSS-induced colitis, and that its effects might be associated with suppression of the TLR4/NF- κ B/NLRP3 inflammasome pathway and GM regulation, which may provide significant supports for the development of potential candidates for UC treatment.

1. Introduction

Ulcerative colitis (UC) is a chronic, colorectal inflammatory disease characterized by mucosal inflammation, abdominal pain, bloody stools, and diarrhea (Høivik et al., 2013). And some studies have pointed out

that UC is the result of various factors (e.g., genetic susceptibility, environmental factors, immune-response disorders, intestinal dysbiosis) (De Souza, 2017), which are related to an increased risk of developing colorectal cancer (Munkholm, 2003). The latter accounts for 15% of deaths in UC patients.

Some chemical and biological agents, such as mesalazine,

* Corresponding author.

** Corresponding author.

*** Corresponding author.

E-mail addresses: 20193096@njucm.edu.cn (Z. Bian), qinyuwen20203090@163.com (Y. Qin), lilin@njucm.edu.cn (L. Li), sll2020@njucm.edu.cn (L. Su), Feichenghao108@163.com (C. Fei), Gene@163.com (Y. Li), 47028239@qq.com (M. Hu), cwx6045@163.com (X. Chen), zhangwei@ahcm.edu.cn (W. Zhang), 300555@njucm.edu.cn (C. Mao), fsyy01080@njucm.edu.cn (X. Yuan), htl2021@njucm.edu.cn (T. Lu), de.ji@njucm.edu.cn (D. Ji).

¹ These authors contributed equally to this work.

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Abbreviations

5-ASA	5-aminosalicylic acid
ASC	apoptosis-associated speck-like protein containing a CARD
CMC-Na	carboxymethylcellulose sodium
DAI	Disease Activity Index
DSS	dextran sulfate sodium
GM	gut microbiota
IL:	interleukin
LefSe	linear discriminant analysis effect size
MDA	malondialdehyde
MPO	myeloperoxidase
OTUs	operational taxonomic units
PCR	thermocycler polymerase chain reaction
PVDF	polyvinylidene fluoride
SC	<i>Schisandra chinensis</i> (Turcz.) Baill
SCE	<i>Schisandra chinensis</i> (Turcz.) Baill. extract
SOD	superoxide dismutase
TNF- α :	tumor necrosis factor- α
UC	ulcerative colitis
ZO	zonula occludens

sulfasalazine, glucocorticoids, immunosuppressive drugs, and anti-TNF- α monoclonal antibody (e.g., adalimumab), have been applied for UC treatment but these agents have various side-effects and limited efficacy (Hoffmann et al., 2019). In contrast, use of herbal medicines shows efficacy and safety in UC treatment (Zhang et al., 2022). Previous study have reported that some natural drugs and herbal medicines exhibit excellent anti-inflammatory ability and efficacy for UC in animal experimental research and clinical trials (Capasso et al., 2014; Guo et al., 2017; Pagano et al., 2016, 2019).

Schisandra chinensis (Turcz.) Baill. (SC) is a plant belonging to family of Magnoliaceae. The dried fruit of SC has been used extensively as a traditional medicinal herb for thousands of years in China. It is also used as traditional tonic in Russia and other countries (Chun et al., 2014; Shikov et al., 2014). It has been used to treat gastrointestinal dysfunction, tumors and liver diseases (Choi et al., 2006; Panossian and Wikman, 2008). Modern pharmacological researches have shown that SC and its related ingredients can reduce inflammation, delay aging and treat tumor (Huang et al., 2008; Kim et al., 2017). Importantly, SC has been used as a traditional herbal medicine for treating diarrhea and recorded in the *Chinese Pharmacopoeia*. Lignan-based compounds isolated from SC, such as schisandrin B and deoxyschisandrin, have been shown to exert anti-colitis effects through suppression of secretion of proinflammatory cytokines and regulation of the nuclear factor-kappa B (NF- κ B) pathway (Liu et al., 2015; Zhang et al., 2016). Furthermore, dibenzocyclooctadiene lignans, the main constituents in SC, are major players in the pharmacodynamic properties of *Schisandra chinensis* (Turcz.) Baill. extract (SCE) (Song et al., 2022; Zhou et al., 2021). Nevertheless, it is important to systematically study the potential mechanism of SCE in the treatment of UC.

The cause and exact pathogenesis of UC are incompletely understood. However, extensive studies have shown that UC is related to the inflammatory response, gut microbiota (GM) disorder, and intestinal immune function (Karakan et al., 2021; Marchesi et al., 2016; Yeşilyurt et al., 2021). Here, we explored the anti-inflammatory effect of SCE by detecting the expression of IL-1 β and other pro-inflammatory cytokines, as well as TLR4, p-p65, NLRP3 and other protein expression in the TLR4/NF- κ B/NLRP3 inflammasome signaling pathway and investigated the GM modulation of SCE in a DSS-induced colitis model in mice, and further interpreted the underlying mechanism of SCE in the treatment of UC.

2. Materials and methods

2.1. Chemicals and reagents

SC (batch number, 190120) was purchased from Nanjing Haiyuan (Nanjing, China). The standards deoxyschisandrin (110764-201714), schisandrin B (110765-201512), schisandrin (110857-201714), and schisantherin A (11529-201706) were obtained from National Institutes for Food and Drug Control (Beijing, China). The standard schizandrol B (F1508140) was bought from Shanghai Aladdin Biochemical Technology (Shanghai, China). The standards gomisins D (M05J9S65019), gomisins J (M22011S128159), schisantherin B (M10D11S133700), schisandrin C (J05GB150952), and angeloylgomisin H (P17J9F65870) were got from Shanghai Yuanye Biological Technology (Shanghai, China). DSS was supplied by Dalian Meilun Biotechnology (Dalian, China). 5-ASA (mesalazine) was purchased from Hengcheng Pharmaceutical Group (Anhui, China). A fecal occult blood test kit was obtained from Beijing Leagene Biotechnology (Beijing, China).

Enzyme-linked immunosorbent assay (ELISA) kits for interleukin (IL)-1 β , TNF- α , IL-18, MPO (myeloperoxidase), malondialdehyde (MDA), and superoxide dismutase (SOD) were purchased from Nanjing JinTing Biotechnology (Nanjing, China). RIPA lysis buffer was bought from Beyotime Biotechnology (Shanghai, China). Protease inhibitor was purchased from Boster Biological Technology (Wuhan, China). Primary antibodies against TLR4, phosphorylated (p)-p65, I κ B- α , phosphorylated (p)-I κ B- α , apoptosis-associated speck-like protein containing a CARD (ASC) were purchased from Nanjing JinTing Biotechnology (Nanjing, China). Primary antibodies against p65 and NLRP3 were purchased from Abcam (Cambridge, UK). Primary antibodies for caspase-1 p20 was got from Proteintech (Rosemont, USA). Primary antibodies for zonula occludens (ZO)-1 and occludin were obtained from Cell Signaling Technology (Danvers, MA, USA). Goat anti-Rabbit secondary antibodies were purchased from MultiSciences (Lianke) Biotech (Hangzhou, China). β -actin was bought from Proteintech.

Acetonitrile of high-performance liquid chromatography (HPLC) grade was purchased from Thermo Fisher Scientific (Waltham, MA, USA). Ultrapure water was purified with a Milli-Q system (Millipore, Milford, MA, USA). Other chemicals and reagents were of analytical grade.

2.2. Preparation of plant extract

SC was formally identified as *Schisandra chinensis* (Turcz.) Baill. by Professor Tu-Lin Lu (Nanjing University of Chinese Medicine, Nanjing, China). An SC voucher specimen (SC-190120) was deposited in the Traditional Chinese Medicine laboratory of Wuxi TCM Hospital Affiliated to Nanjing University of Chinese Medicine under closed and dry storage conditions at room temperature.

After the dried SC fruits were crushed, a certain amount of sample was extracted with 8 vol of 85% alcohol for 2 h under a reflux condenser. The extraction was repeated two further times. The filtrates were combined, concentrated and dried with a rotary evaporator (Su et al., 2018). The yield of SC alcohol extract was 30.45% (w/w). SCE (5.3 mg) was weighed and mixed with 5 mL of methanol. This mixture underwent ultrasonication (300 W, 40 kHz) for 10 min at room temperature. Then, the supernatant was filtered through a 0.22 μ m membrane to obtain prepared samples for HPLC. SCE was analyzed by a 1200 HPLC system (Agilent Technologies, Santa Clara, CA, USA) equipped with a diode array detector from the same manufacturer. Chromatographic separation was accomplished in an Eclipse XDB-C18 (250 mm $\times$ 4.6 mm, 5 μ m) column (Agilent Technologies) maintained at 30 $^{\circ}$ C. The mobile phase was 0.2% (v/v) phosphoric acid (A) and acetonitrile (B). The gradient elution was as follows: 0–20 min, 40–60% B; 20–35 min, 60–68% B; 35–45 min, 68–78% B; 45–55 min, and 78–40% B. When the detection wavelength was set at 217 nm, the flow rate was 1 mL/min, and 10 μ L of each sample was injected after equilibration.

2.3. Animal experimentation

Male C57BL/6 mice (20 ± 2 g) were obtained from Hangzhou Medical College (Permission No: SCXK (ZHE) 2019-0002). Mice were housed in a breeding environment (25 ± 2 °C, 60 ± 10% relative humidity, and a periodic cycle of 12-h light/darkness) and had free access to water and food. Procedures involving animal experimentation were approved by the Ethics Committee of Wuxi TCM Hospital Affiliated to Nanjing University of Chinese Medicine (Approval ID: SWJWDW2021102901) and conducted in accordance with the guidelines for laboratory animals care and use of the National Institutes of Health.

Mice were allowed to acclimatize to their environment for 7 days. Then, they were divided randomly into five groups of eight: normal, DSS, 5-ASA (5-ASA + DSS), SCEL (low-dose SCE + DSS) and SCEH (high-dose SCE + DSS). The normal group received physiological (0.9%) saline without DSS. The other mice received 3% (w/v) DSS in drinking water for 7 days (Vlantis et al., 2016). On day-8, the SCEL and SCEH group were administered SCE (dissolved in 0.5% carboxymethylcellulose sodium (CMC-Na) solution) at a dose of 593.78 mg/kg (equivalent to a crude drug dose of 1.95 g/kg, 2.5-times the human-equivalent dose) and 1187.55 mg/kg (equivalent to a crude drug dose of 3.90 g/kg, 5-times the human-equivalent dose) once a day via the oral route for 14 days, respectively (Ahn et al., 2015; Food and Drug Administration, 2005). The 5-ASA (positive treatment) group was administered 300 mg/kg of 5-ASA (dissolved in 0.5% CMC-Na solution) once a day via the oral route for 14 days. The other two groups of experimental mice were treated with equal volume of 0.5% CMC Na solution. At the end of the experiment, all mice were killed using CO₂. Colon tissues and intestinal contents were harvested rapidly.

2.4. Evaluation of colitis

The body weight and Disease Activity Index (DAI) scores of mice were recorded every day (Wang et al., 2019). Mice were monitored every day to determine weight and Disease Activity Index (DAI) score. The DAI score criteria included the following: bodyweight loss (0, unchanged; 1, 1–5%; 2, 5–10%; 3, 10–15%; 4, >15%), stool consistency (0, “normal”; 2, “loose” stools; 4, “diarrhea”), and occult/gross blood (0, “normal”; 2, “hemocult positive”; 4, “gross bleeding”). Divide the total score of the three parameters stated above by 3 to get the DAI score. The colon length was measured immediately after mice had been killed. The colon tissues were quickly taken out and soaked in 4% paraformaldehyde and embedded in paraffin. Sections (4 µm thick) were deparaffinized, and then stained with hematoxylin and eosin (H&E). Finally, colon lesions were observed by microscopy and the histopathology score was determined according to our previous report (Su et al., 2020).

2.5. Measurement of cytokines and oxidative stress indexes levels in the colon

After thawing, the colon tissue were mixed with ice-cold phosphate buffered saline (w/v, 1:9) and homogenized with a tissue homogenizer. The samples were centrifuged at 613×g for 10 min at 4 °C and the supernatant was used for analysis. The activity of MPO, MDA, and SOD, and the level of IL-1β, TNF-α, and IL-18, in the above samples were measured with ELISA kits according to kit instructions. The total protein of each sample was detected with a BCA protein assay kit (Pierce, Rockford, IL, USA).

2.6. Western blotting

RIPA buffer containing phenylmethanesulfonyl fluoride and protease inhibitor was added to the colon tissue, and then homogenized. The extract was determined with the BCA protein assay kit (Bestbio, Shanghai, China). Equal amounts of protein (50 µg) were separated by

sodium dodecyl sulfate–polyacrylamide gel electrophoresis and moved to polyvinylidene fluoride (PVDF) membranes (Merck Millipore, Whitehouse Station, NJ, USA). Then, the PVDF membranes were blocked in 5% nonfat dried milk for 2 h and incubated with primary antibodies against TLR4, NF-κB p65, p-p65, IκB-α, p-IκB-α, NLRP3, ASC, caspase-1 p20, ZO-1, and occludin, all at 1:1500 dilution, overnight at 4 °C. Afterwards, the PVDF membranes were incubated with secondary antibody (1:2000 dilution) for 2 h at room temperature. Protein bands were detected by ImageJ (US National Institutes of Health, Bethesda, MD, USA).

2.7. 16s rRNA analyses of intestinal contents

Genomic DNA was extracted from intestinal contents using an OMEGA-soil DNA kit (Omega Bio-tek, Norcross, GA, USA). The V3–V4 hypervariable region of the 16s rRNA gene of bacteria was amplified by a thermocycler polymerase chain reaction (PCR) (GeneAmp 9700; Applied Biosystems, Foster City, CA, USA) using the forward 338 (5'-ACTCTACGGGAGGCAGCAG-3') and reverse 806 (5'-GGACTACHVGGGTWTCTAAT-3'). PCR products were purified by the AxyPrep DNA Gel Extraction Kit (Axygen Biosciences, Union City, CA, USA) and quantified by QuantiFluor-ST (Promega, Fitchburg, WI, USA) in accordance with Kit instructions. Purified amplicons were pooled in equimolar and paired-end sequences (2 × 300) using the MiSeq™ platform (Illumina, San Diego, CA, USA). Bioinformatics analysis of raw sequence data was done using the QIIME2 platform (version 2.4). Demultiplexed sequences from each sample were quality-filtered, denoised, and merged, and the chimeric sequences were removed. A taxonomy table was generated from the 99% pretrained Greengenes database (version 13-8; <https://greengenes.secondgenome.com/>) by the QIIME2 feature-classifier plugin. Analyses of alpha and beta diversity were undertaken using the core-diversity plugin within QIIME2. Columnar stacking diagrams, heatmaps, and linear discriminant analysis effect size (LEfSe) were used to screen the abundance of bacterial communities among each group using Wekemo Bioincloud (www.bioincloud.tech/). A sequencing service was provided by Wekemo Tech Group (Shenzhen, China).

2.8. Statistical analyses

Experimental data were shown as the mean ± Standard deviation (SD) and were analyzed by one-way analysis of variance followed by Dunnett's test using Prism 8 (GraphPad, San Diego, CA, USA). *P* < 0.05 was considered significant.

3. Results

3.1. SCE components

SCE components were analyzed by HPLC. Ten lignans were determined by their retention time and ultraviolet spectrum to the reference standards. These were schisandrin, gomisin D, gomisin J, schizandrol B, angeloylgomisin H, schisantherin A, schisantherin B, deoxyschisandrin, schisandrin B and schisandrin C, and their content was 12.27, 0.69, 0.74, 4.20, 2.63, 0.57, 2.53, 3.16, 6.26, and 0.94 mg/g, respectively (Fig. 1A and B). Dibenzocyclooctene lignans were the major substances in SCE.

3.2. SCE alleviated DSS-induced colitis

The detailed animal-experimental procedure is shown in Fig. 2A. Compared with normal mice, DSS-treated mice showed significant loss in bodyweight, while the weight of each treatment group (SCEL, SCEH and 5-ASA) recovered significantly compared with DSS group (Fig. 2B). Compared with normal mice, the DAI score of the DSS group was markedly increased (*P* < 0.01) (Fig. 2C), and low and high doses groups of SCE or 5-ASA decreased the DAI score significantly and alleviated

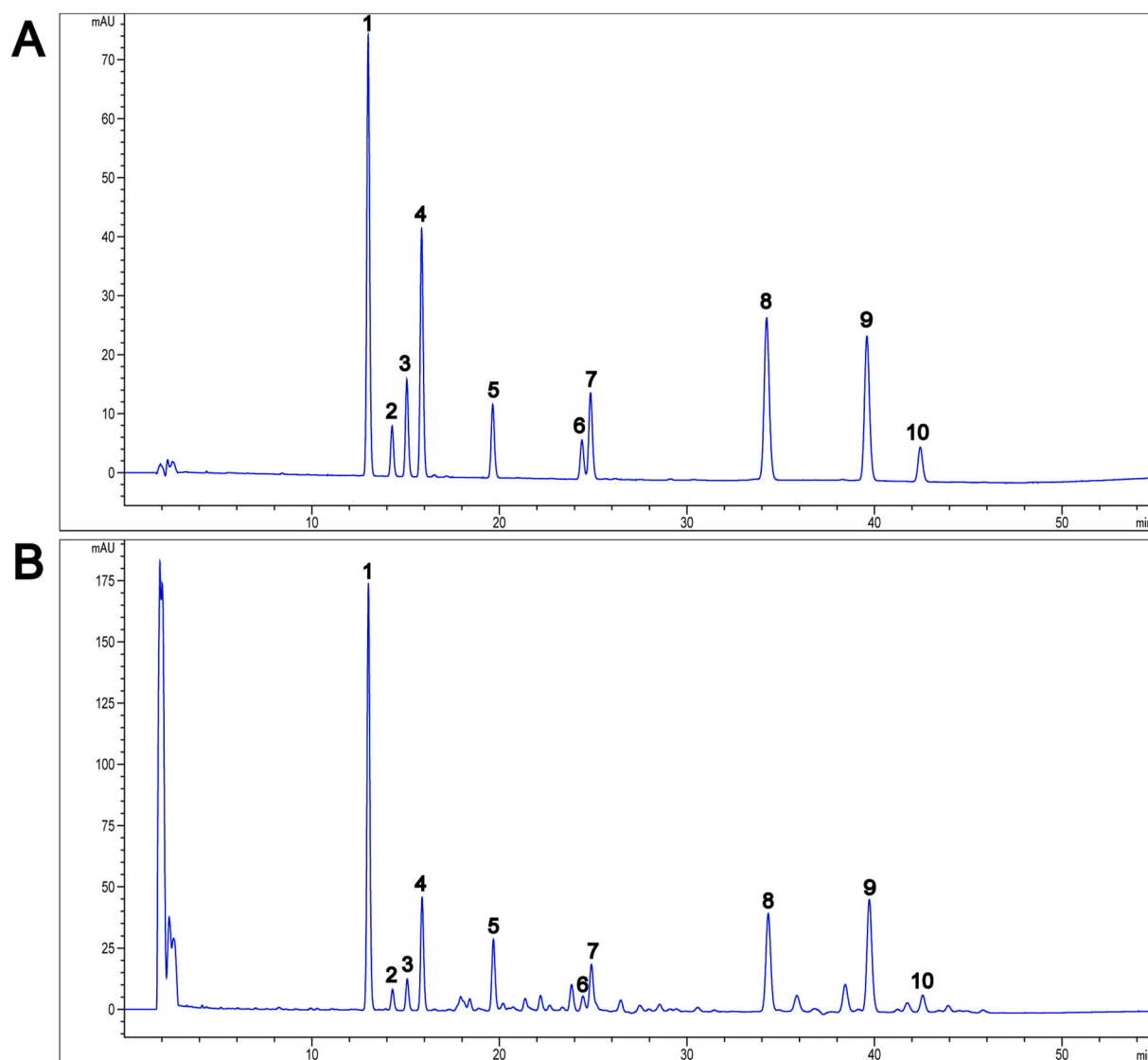


Fig. 1. High-performance liquid chromatography (HPLC) chromatogram of reference standards (A) and SCE (B). Ten lignans in SCE were schisandrin (1), gomisin D (2), gomisin J (3), schizandrol B (4), angeloylgomisin H (5), schisantherin A (6), schisantherin B (7), deoxyschisandrin (8), schisandrin B (9), and schisandrin C (10), respectively.

colitis symptoms compared with the UC disease model mice ($P < 0.01$). DSS treatment shortened the colon length significantly ($P < 0.01$), whereas low and high doses of SCE or 5-ASA inhibited DSS-induced colon shortening significantly ($P < 0.01$) (Fig. 2D and E). Mice treated with DSS suffered complete destruction of tissue structure (Fig. 2F and G) as manifested by massive infiltration of inflammatory cells, distortion/loss of crypt structure, and abscesses of goblet cells. Low and high doses of SCE administration led to reduction of histologic changes, as evidenced by a decrease in aggregation of inflammatory cells, accompanied by restoration of crypt structure and goblet cells. Histopathology showed that DSS treatment induced an increase in the histology score, whereas SCE administration decreased the histology score.

3.3. Changes in levels of cytokines and oxidative stress indexes

The levels of cytokines in the colon were determined to evaluate the anti-inflammatory effect of SCE in DSS-induced colitis (Fig. 3A–F). The activity of MPO and MDA, and levels of IL-1 β , TNF- α , and IL-18, in mice of the DSS group were increased significantly as compared with those in the normal group ($P < 0.01$), whereas the SOD activity was decreased ($P < 0.01$). Compared with the DSS group, administration of low and high

doses of SCE or 5-ASA inhibited the activity of MPO and MDA, levels of IL-1 β , TNF- α , and IL-18 ($P < 0.01$), and increased the activity of SOD ($P < 0.01$). The results stated above demonstrated that SCE had anti-inflammatory effects in DSS-induced colitis in mice by inhibiting secretion of proinflammatory cytokines such as IL-1 β , TNF- α , and IL-18. SCE improved the effect of oxidative stress in DSS-induced colitis in mice by decreasing the activity of MPO and MDA, and increasing the activity of SOD.

3.4. Suppression of the TLR4/NF- κ B signaling pathway

The TLR4/NF- κ B signaling pathway has a vital role in regulating the production of cytokines. Hence, we measured expression of crucial proteins in the TLR4/NF- κ B signaling pathway (Fig. 4A–D). Contents of TLR4, p-p65, and p-I κ B- α were increased significantly upon DSS intake ($P < 0.001$). High dose of SCE administration suppressed expression of TLR4, p-p65, and p-I κ B- α ($P < 0.01$), thereby resulting in inhibition of the TLR4/NF- κ B signaling pathway.

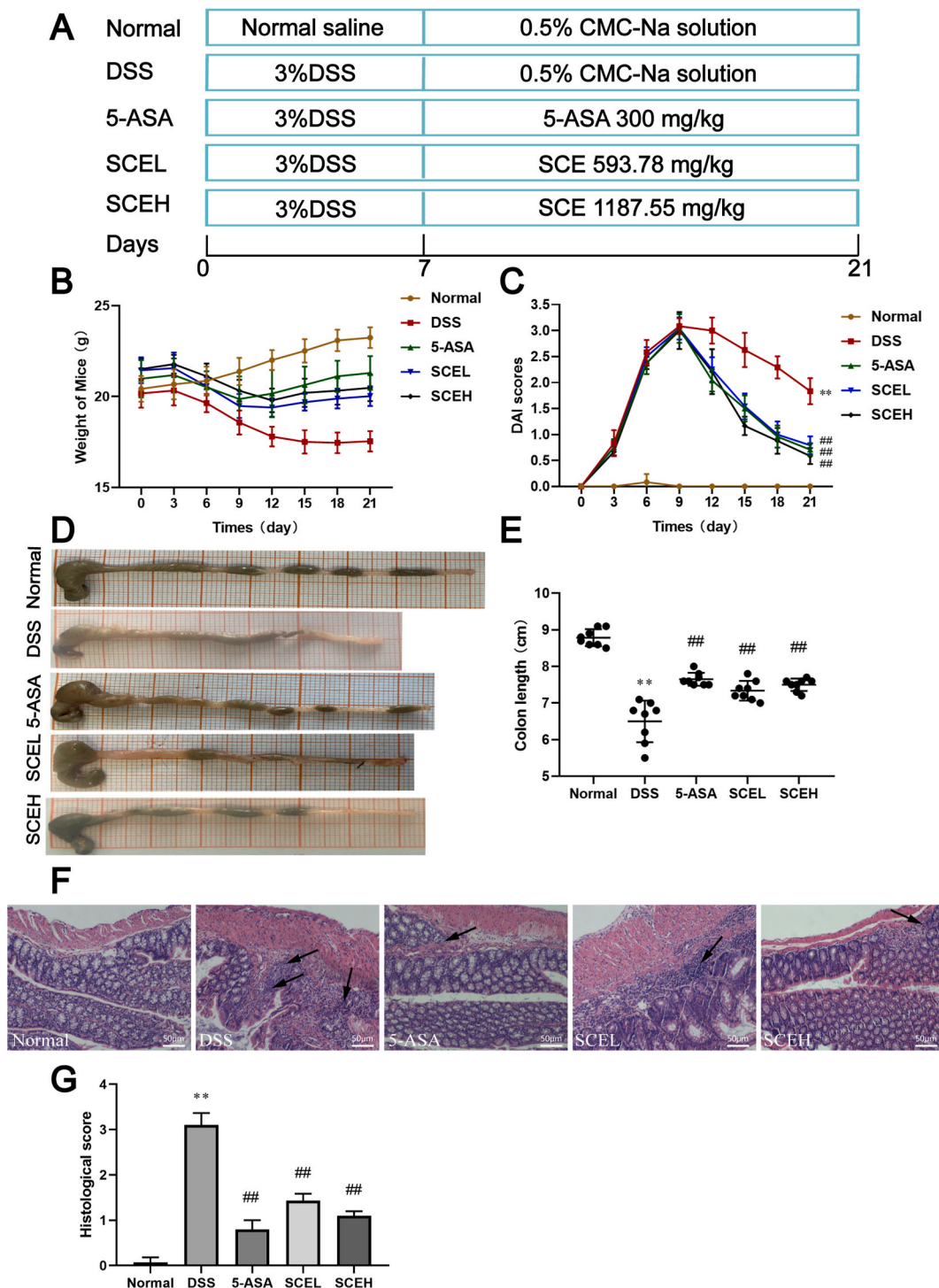


Fig. 2. SCE ameliorated DSS-induced colitis in mice. (A) Detailed animal experimental procedure. (B) Bodyweight changes during UC progression. (C) DAI score of each group. (D) Representative colon images from each group. (E) Colon length of mice from different groups. (F) Hematoxylin and eosin (H&E) staining of colon tissues at $\times 200$ magnification. Arrows: the infiltration of inflammatory cells. (G) Histopathology score of colon damage. Normal, normal group; DSS, DSS-induced colitis model group; 5-ASA, 5-ASA-positive group; SCEL, low-dose SCE group; SCEH, high-dose SCE group. $**P < 0.01$ compared with the normal group, $##P < 0.01$ compared with the DSS group. Data are the mean $\pm$ SD. $n = 8$.

3.5. Inhibition of activation of the NLRP3 inflammasome pathway

We also evaluated if SCE prevented DSS-induced colon damage by regulating activation of the NLRP3 inflammasome pathway. DSS treatment increased expression of NLRP3, ASC, and caspase-1 p20 significantly compared with that in the control group ($P < 0.001$) (Fig. 5A–D). Conversely, high dose of SCE treatment downregulated expression of

NLRP3 ($P < 0.01$), ASC ($P < 0.01$), and caspase-1 p20 ($P < 0.001$) significantly. The above results showed that SCE could prevent or treat UC by inhibiting the activation of NLRP3 inflammatory body pathway.

3.6. Regulation of expression of ZO-1 and occludin

SCE ameliorated DSS-induced histopathologic changes, so we

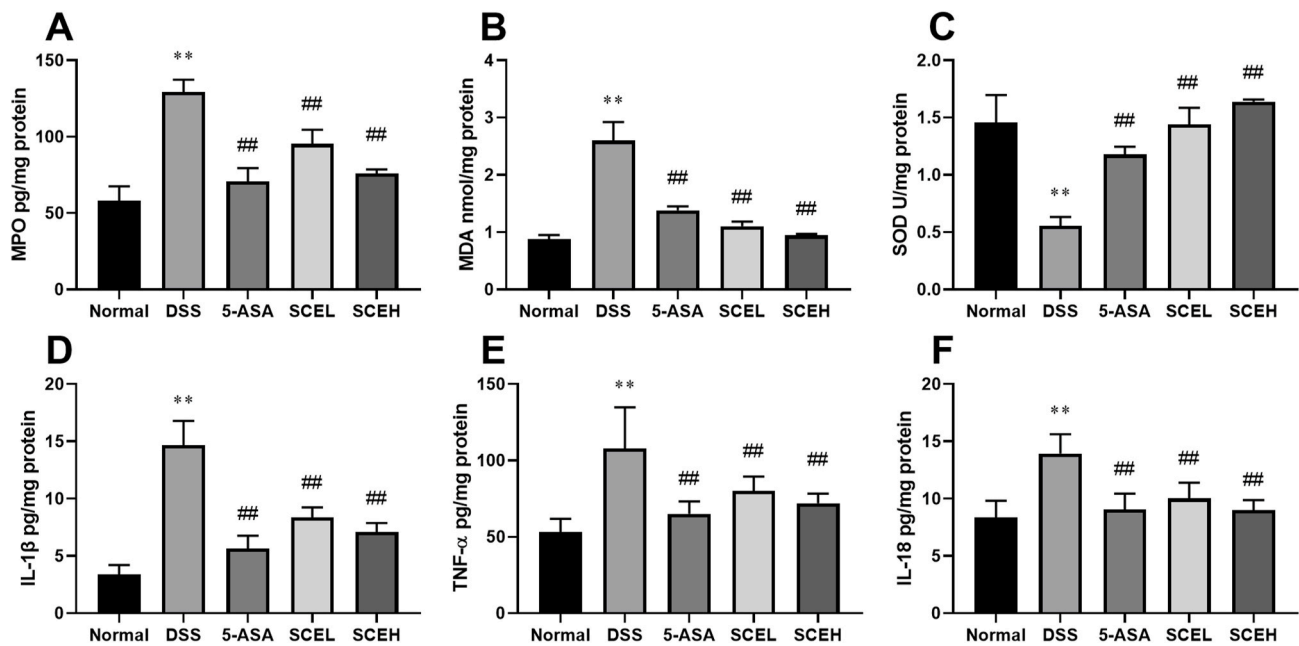


Fig. 3. SCE effects on UC mice. Activity of MPO (A), MDA (B), and SOD (C), and levels of IL-1 β (D), TNF- α (E), and IL-18 (F) of colon tissues were determined by ELISAs. ** $P < 0.01$ compared with the normal group, ## $P < 0.01$ compared with the DSS group. Data are the mean $\pm$ SD. $n = 8$.

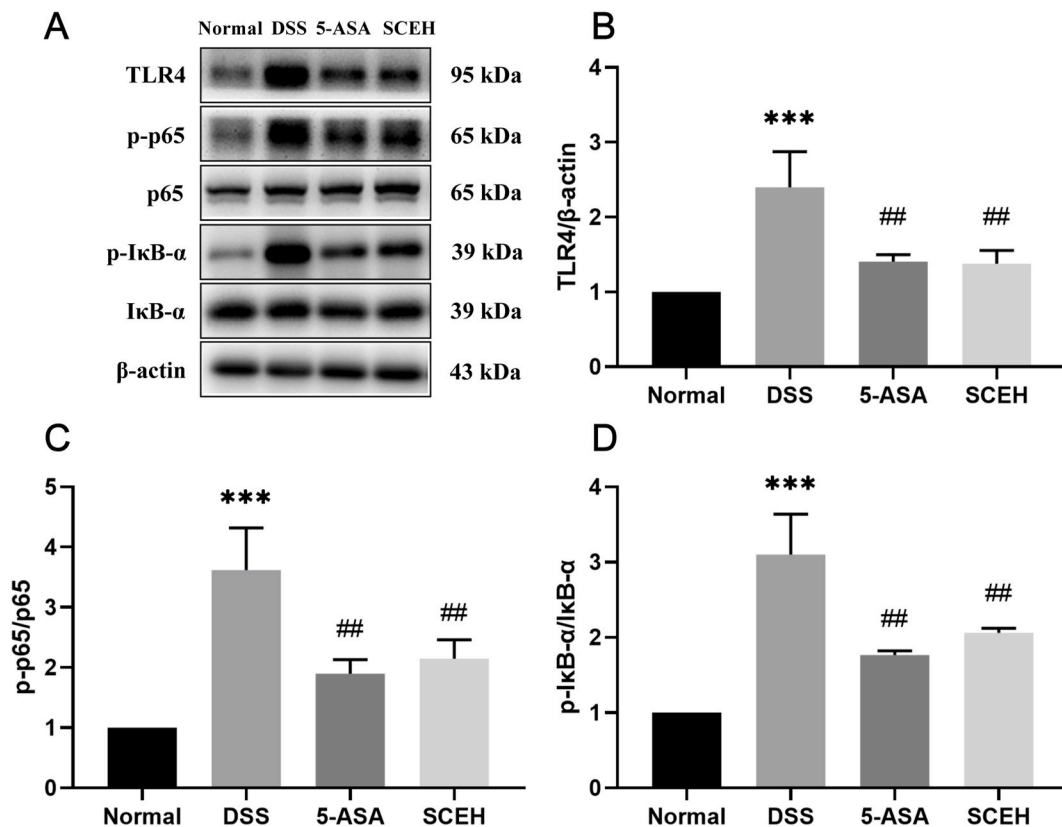


Fig. 4. SCE effects on regulation of the TLR4/NF- κ B signaling pathway in mice with DSS-induced colitis (A). Expression of TLR4 (B), p-p65 (C), and p-I κ B- α (D) were assessed by western blotting in mice with DSS-induced colitis. *** $P < 0.01$ compared with the normal group, # $P < 0.05$, ## $P < 0.01$ compared with the DSS group. Data are the mean $\pm$ SD. $n = 3$.

speculated that SCE might have a regulatory effect on DSS-induced expression of intestinal tight junction (TJ) proteins. Expression of ZO-1 and occludin was suppressed significantly after DSS intake compared with that in the normal group ($P < 0.001$, $P < 0.05$)

(Fig. 6A–C). However, high dose of SCE treatment upregulated protein expression of ZO-1 significantly in comparison with that in the DSS group ($P < 0.01$), and expression of occludin showed an upward trend after the intervention of SCE.

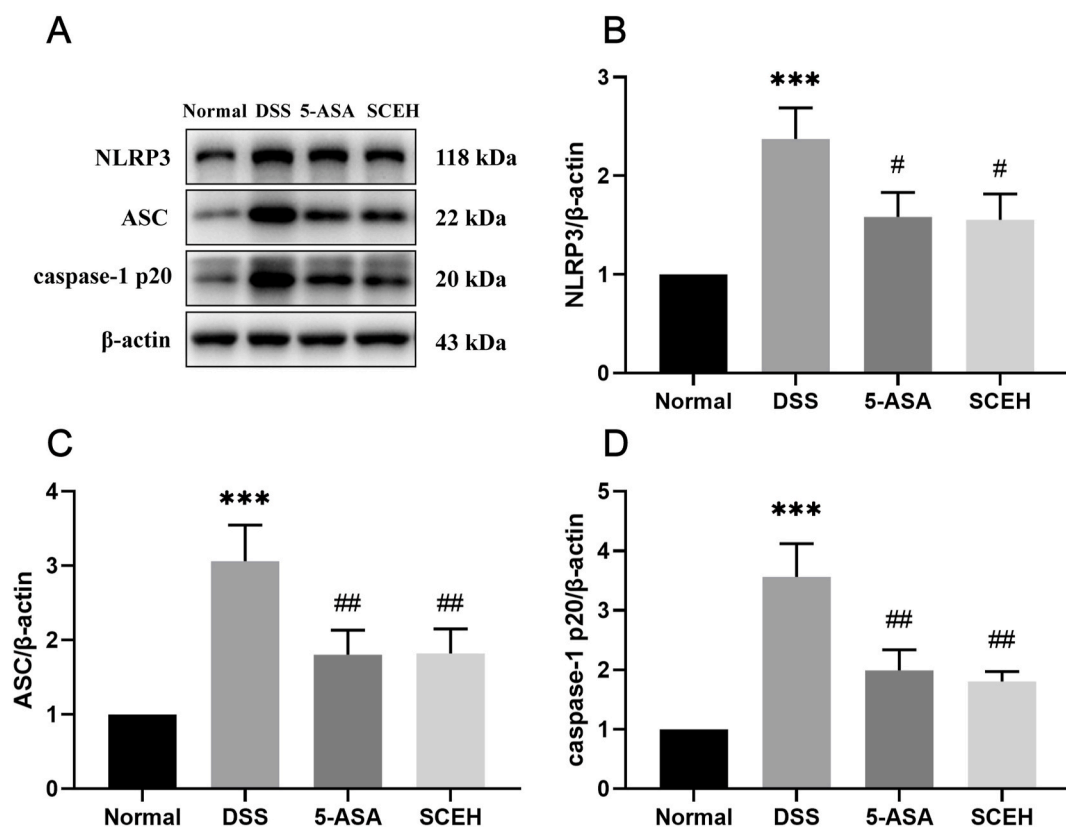


Fig. 5. SCE effects on regulation of the NLRP3 inflammasome signaling pathway in mice with DSS-induced colitis (A). Protein expression of NLRP3 (B), ASC (C), and caspase-1 p20 (D) was determined by western blotting in mice with DSS-induced colitis. $^{**}P < 0.01$ compared with the normal group, $^{\#}P < 0.05$, $^{\#\#}P < 0.01$ compared with the DSS group. Data are the mean $\pm$ SD. $n = 3$.

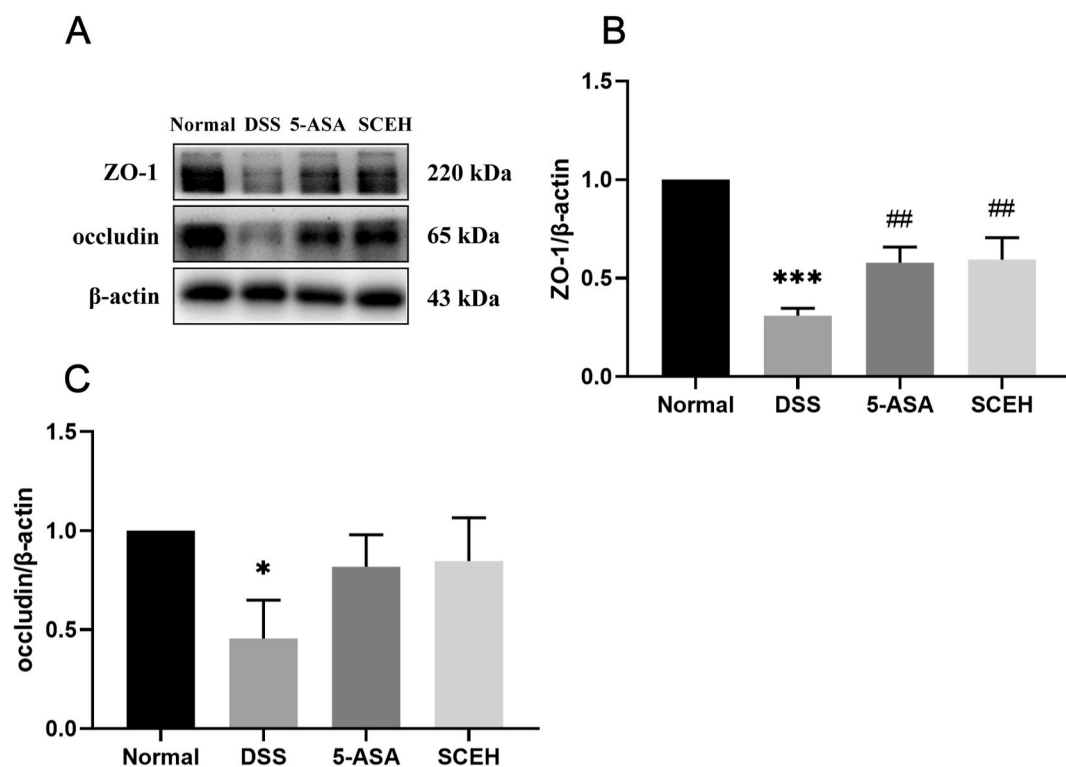


Fig. 6. SCE effects on regulation of intestinal tight junction (TJ) proteins of colon tissues in mice with DSS-induced colitis (A). Expression of ZO-1 (B) and occludin (C) was determined by western blotting. $^{**}P < 0.01$ compared with the normal group, $^{\#}P < 0.05$, $^{\#\#}P < 0.01$ compared with the DSS group. Data are the mean $\pm$ SD. $n = 3$.

3.7. Effect of SCE on the diversity and composition of the GM

3.7.1. Alpha and beta diversity of the GM

Recent studies have suggested that GM imbalance can result in UC development, and that modulation of intestinal flora can ameliorate UC (Li et al., 2020; Sun et al., 2020). Therefore, we investigated if the ameliorative effects of SCE were related to regulation of intestinal microbiota. Some *plateaued* rarefaction curves of operational taxonomic units (OTUs) indicated that the quality of sequencing was good (Fig. 7A and B). Analyses of alpha diversity showed that DSS did not affect the diversity of microbial species significantly (Shannon Index), but showed

a decreasing trend (Fig. 7C). The community richness (Faith's Phylogenetic Diversity Index) in the UC model group was decreased markedly when compared with the normal group ($P < 0.05$), and low dose of SCE could prevent the decreasing effect of DSS significantly ($P < 0.05$) (Fig. 7D). Principal coordinates analysis (PCoA) and nonmetric multi-dimensional scaling (NMDS) were undertaken to reflect beta diversity (Fig. 7E and F). PCoA and NMDS demonstrated that DSS treatment disturbed the overall structure of intestinal flora markedly, and an overall structural shift of intestinal flora towards the normal group was observed upon SCE treatment.

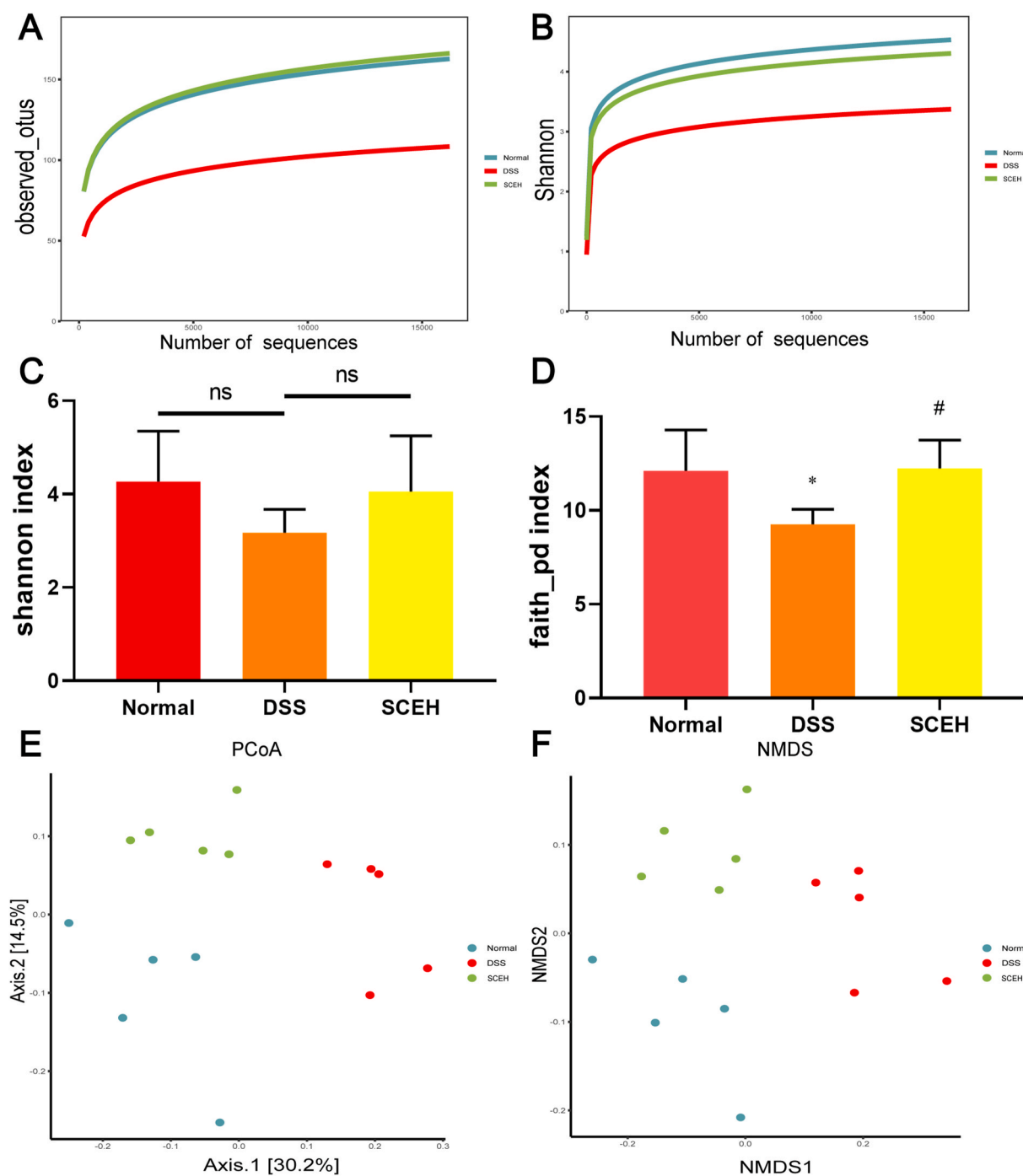


Fig. 7. SCE effects on structural modulation of the gut microbiota in mice with DSS-induced colitis. (A) The rarefaction curve of OTUs. (B) The rarefaction curve of each group using the Shannon Index. Analyses of the Shannon Index (C) and Faith's Phylogenetic Index (D) for alpha diversity. PCoA (E) and NMDS (F) for analyses of beta diversity. * $P < 0.05$ compared with the normal group, # $P < 0.05$ compared with the DSS group, ns, no significant difference. Data are the mean $\pm$ SD. $n = 5$.

3.7.2. GM composition

GM composition at phylum and genus levels in different groups is shown in Fig. 8. Classified by phylum level, microbial community of all experimental mice mainly contained Firmicutes, Bacteroidetes, Actinobacteria, Proteobacteria, and Verrucomicrobia (Fig. 8A), and the relative abundance of bacteria from each group has been shown in Supplementary Table S1. There was a remarkable decrease in the relative abundance of bacteria from Bacteroidetes, but a significant increase in bacteria from Actinobacteria in the UC disease model mice compared with healthy mice. However, low dose of SCE intervention increased the relative abundance of bacteria from Bacteroidetes significantly, and decreased the relative abundance of bacteria from Actinobacteria, compared with that in the DSS group. The community composition of microbiota at the genus level was determined by heatmaps (Fig. 8B), and the relative abundance of bacteria from each group has been shown in Supplementary Table S2. The relative abundance of bacteria classified by genera level *Turicibacter*, *Ruminococcus*, *Odoribacter*, *Oscillospira*, and *Parabacteroides* was decreased significantly, whereas the abundance of bacteria of the genera *Streptococcus* and *Adlercreutzia* was increased obviously, after DSS treatment. These shifts were reversed upon SCE treatment.

LEfSe was employed to identify the dominant microbiota in five groups. First, the UC disease model group was compared with healthy mice (Fig. 8C and D). Classified by phylum level, the contribution of bacteria from Actinobacteria in the UC disease model group was greater, and the contribution of bacteria from Bacteroidetes and Verrucomicrobia in healthy mice was greater. Classified by genus level, the contribution of *Streptococcus* in the UC disease model group was greater, and the contribution of *Turicibacter* and *Akkermansia* in the healthy mice was greater. Second, the SCE low-dose treatment group was compared with the UC disease model group (Fig. 8E and F). In the SCE low-dose treatment group, the abundance of bacteria from the phylum Proteobacteria and genus *Turicibacter* was increased significantly. The abundance of bacteria of the genus *Streptococcus* was higher in the DSS group.

We wished to investigate if changes in the GM in mice suffering from colitis were correlated with UC parameters. Hence, the relationship between changes in the GM and the DAI score, colon length, histology score, activity of MPO, MDA, and SOD, and levels of IL-1 β , TNF- α , and IL-18 were analyzed by Spearman's correlation analysis (Fig. 9). Bacteria of the phylum Actinobacteria and genus *Streptococcus* were negatively correlated with parameters that prevent UC. Bacteria of the genera *Parabacteroides*, *Odoribacter*, *Turicibacter*, *Oscillospira*, and *Ruminococcus* were negatively correlated with parameters that promote UC. These results indicated that proinflammatory cytokines may act on the GM which is associated with colitis. The mechanism of SCE-ameliorated intestinal inflammation might be the suppression of secretion of proinflammatory cytokines, the regulation on the immune micro-environment and intestinal micro-ecology.

4. Discussion

Research indicates that the pathogenic mechanism of UC is related to dysregulation of the inflammatory response, oxidative stress, and intestinal microecology of the host (Castro-Dopico et al., 2019; Galipeau et al., 2021; Ungaro et al., 2017). Despite use of chemical and biological agents, UC is prone to relapse, which can lead to high medical expenses for patients. There is an urgent need to find alternative drugs for UC treatment.

SC has historically been ascribed to be a well established tonic or adaptogen to increase the resistance to disease (Panossian et al., 2021). As a TCM formulation used to treat diarrhea, SC has also been reported to have anti-inflammatory and antioxidant properties (Kopustinskiene and Bernatoniene, 2021; Park et al., 2021). Here, we demonstrated that SCE could alleviate DSS-induced colitis symptoms and inhibit secretion of proinflammatory cytokines.

We employed a colitis model used widely in which symptoms closely

resemble the clinical manifestations of UC in humans (Li et al., 2021). Ingestion of DSS causes chemical damage to the intestinal epithelium, exposing the lamina propria to lumen antigens and intestinal bacteria, leading to intestinal immune and inflammatory response (Bilsborough et al., 2021). Bodyweight loss, DAI score, and shortened colon length are usually considered as inflammatory indicators when evaluating UC progression (Li et al., 2019; Yu et al., 2020). Besides, histopathologic changes in patients with UC usually include infiltration of inflammatory cells, loss of crypt structure, and an absence of goblet cells in colonic tissue (McGuckin et al., 2009). Herein, SCE pretreatment ameliorated the symptoms of colitis significantly, as evidenced by bodyweight gain, reduction in the DAI score, restored colon length, and alleviated histopathologic changes. It has been reported that MPO is an important inflammatory marker related to infiltration of neutrophils and tissue damage, and is expressed abundantly in the colonic tissues of UC patients (Chami et al., 2018). Consistent with other studies, MPO concentration was increased markedly in the colonic tissues of mice with DSS-induced colitis, whereas intervention with SCE reduced MPO concentration significantly. MDA and SOD are important markers of oxidative stress closely associated with the onset and progression of UC (Wang et al., 2021). We revealed that MDA concentration improved in the colon of mice with DSS-induced UC, whereas the activity of SOD (an important antioxidant enzyme) was reduced markedly. However, SCE treatment decreased MDA concentration significantly and enhanced SOD activity. Increasing evidence has shown that inflammatory over-reaction and activation of proinflammatory cytokines are implicated in UC pathogenesis (Zhu et al., 2021). Excessive production and accumulation of proinflammatory cytokines such as IL-1 β , TNF- α , and IL-18 can trigger an inflammatory resulting in colon damage during colitis progression. IL-1 β is secreted by macrophages and exhibits a proinflammatory effects associated with the UC activity. TNF- α is an important cell factor for leading to the destruction of colonic mucosal barrier in UC, and inhibition of TNF- α expression is an effective way to treat UC. IL-18 is a proinflammatory factor, and its expression and activity is related to UC severity (Wang et al., 2014). IL-18 participates in damage to mucosal barrier function and stimulates inflammation during the UC course (Williams et al., 2019). The results of this study show that the level of the proinflammatory cytokines IL-1 β , TNF- α and IL-18 in mice suffering from colitis was suppressed markedly by SCE treatment. These results suggest that SCE improves oxidative stress and ameliorates the inflammatory response in DSS-induced colitis.

Growing evidence has suggested that the inflammatory response is modulated primarily by the TLR4/NF- κ B signaling pathway, which are directly related to the pathogenesis of UC (Zhou et al., 2017). TLR4 is an important member of the TLR family, which are transmembrane proteins. TLR4 shows high expression in the colonic tissues of patients and animals suffering from UC (Chamanara et al., 2019; Murad, 2014). In the pathological conditions of UC, lipopolysaccharide released from Gram-negative bacteria activates host inflammatory responses by acting on the TLR4 receptor (Luo et al., 2020). This action triggers activation of the downstream NF- κ B pathway and production of proinflammatory mediators that contribute to development of intestinal inflammation in UC. NF- κ B activation is associated with interaction with I κ B. Through phosphorylation, NF- κ B p65 can be released from the p65-p50-I κ B complex. This phenomenon is followed by nuclear translocation and resultant production of cytokines such as IL-1 β , TNF- α and IL-6 and contribution to the inflammatory response (Chen et al., 2020; Kim et al., 2006). Therefore, we investigated the regulatory efficacy of SCE on the TLR4/NF- κ B pathway in mice with DSS-induced colitis. We showed that DSS administration upregulated expression of TLR4, p-p65, and p-I κ B- α markedly, and increased the level of IL-1 β , TNF- α , and IL-18. SCE treatment significantly decreased protein expression of TLR4, p-p65, and p-I κ B- α and downstream proinflammatory cytokines in mice with DSS-induced colitis. These results suggest that the anti-inflammatory efficacy of SCE on UC is based on inhibition of the TLR4/NF- κ B signaling pathway.

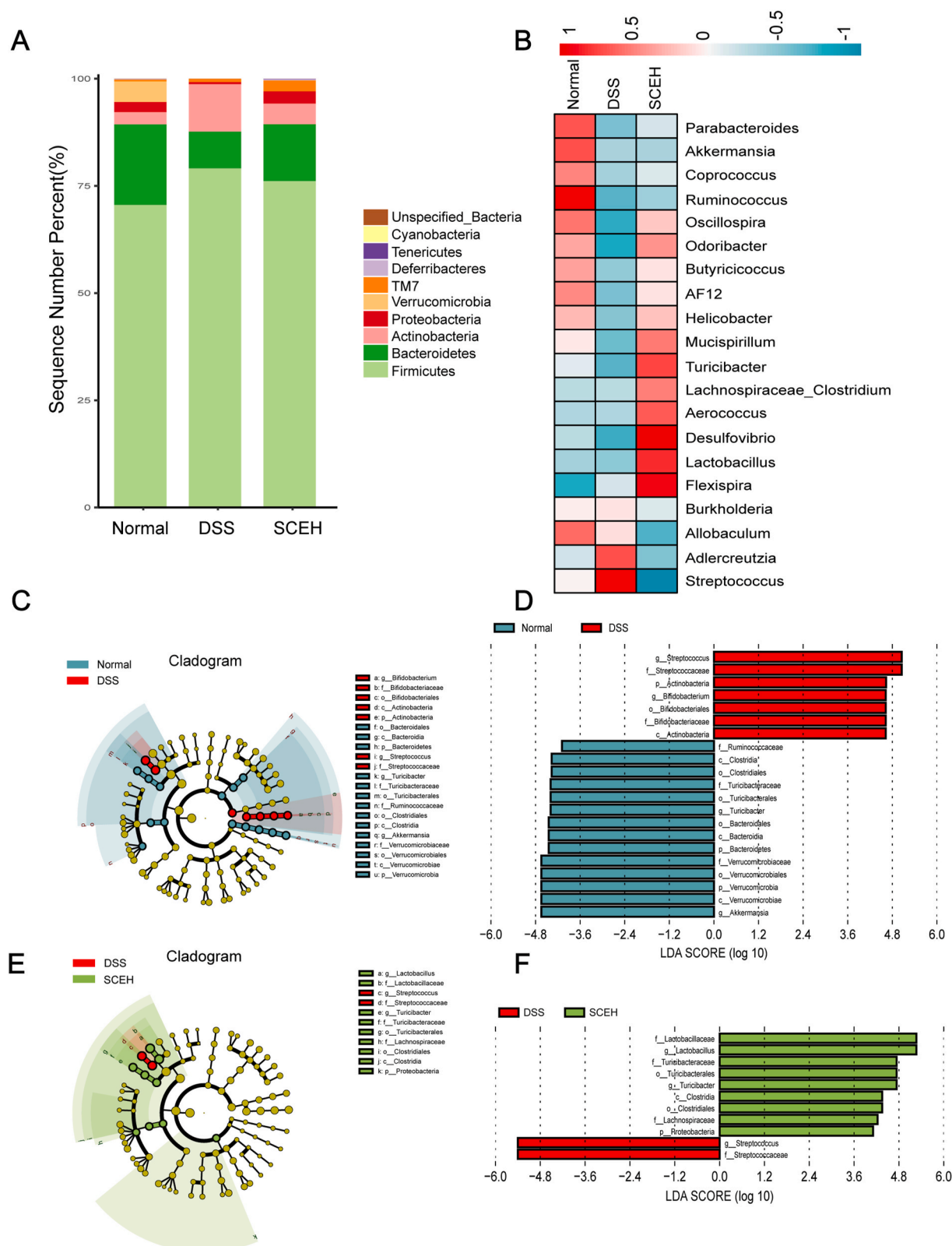


Fig. 8. SCE effects on the community composition of the gut microbiota in mice with DSS-induced colitis. (A) Columnar stacking diagram of the microbial-community composition at the phylum level. (B) Heatmap of the microbial community-composition at the genus level. (C) Cladogram obtained from LEfSe between the normal group and DSS group. (D) LDA scores indicating the bacterial taxa enriched significantly in the gut microbiota between the normal group and DSS group. (E) Cladogram obtained from LEfSe between the DSS group and SCEH group. (F) LDA score indicating the bacterial taxa enriched significantly in the gut microbiota between the DSS group and SCEH group. LDA-score threshold of bacterial taxa was set to >4 and $n = 5$.

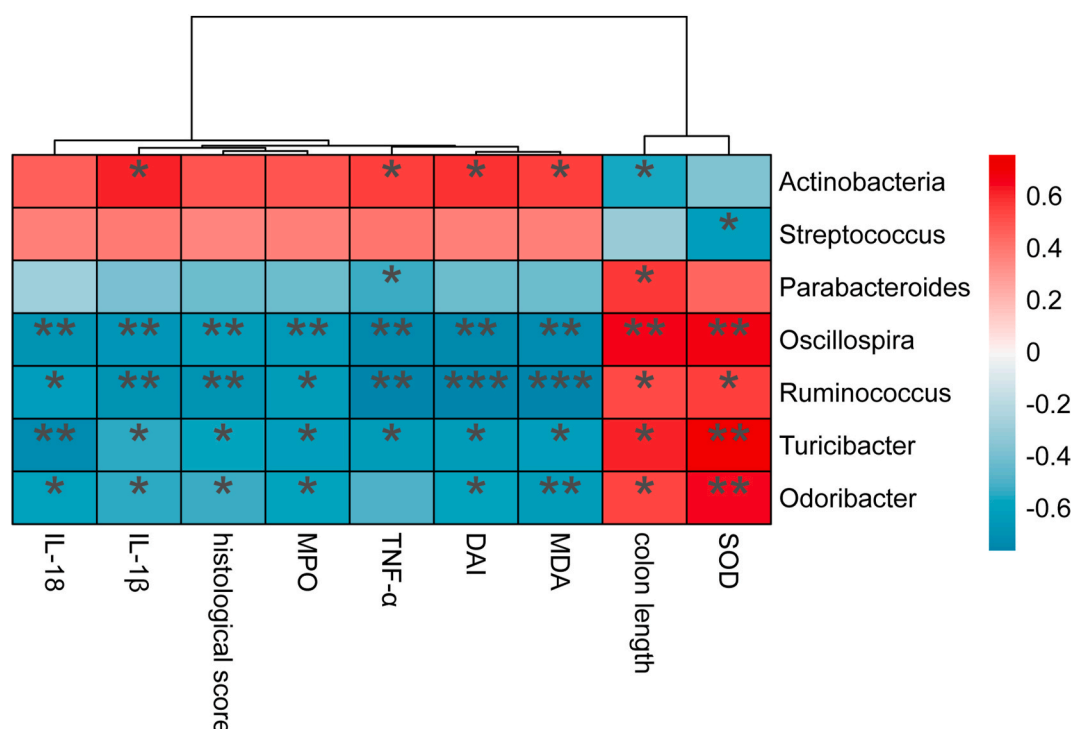


Fig. 9. Heatmap of Spearman's correlation analysis between the changes in the gut microbiota and UC parameters. Red and blue represent positive and negative correlations, respectively. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Secretion of proinflammatory cytokines such as IL-1β and IL-18 requires inflammasome activation (Zhen and Zhang, 2019). The NLRP3 inflammasome, one of nucleotide-binding and oligomerization domain-like receptors (NLRs) family, functions as a “molecular platform” participating in progression of various inflammatory diseases, including UC (Mai et al., 2019). The activated NLRP3 inflammasome unites with ASC to stimulate pro-caspase-1. This action leads to production of active caspase-1, which promotes the maturation and release of the proinflammatory cytokines IL-1β and IL-18 (Lu et al., 2014). We showed that protein expression of the NLRP3 inflammasome, ASC, and caspase-1 was upregulated significantly in DSS-induced UC mice, which was also associated with an increase of the level of IL-1β and IL-18 in colonic tissues. Conversely, SCE administration suppressed activation of the NLRP3 inflammasome significantly. These data indicated that SCE alleviated UC, which may be mediated (at least in part) through suppression of activation of the NLRP3 inflammasome pathway.

Monolayered columnar epithelial cells are tightly connected by TJ proteins, which are indispensable physical barriers used to maintain intestinal integrity (Yu et al., 2018). Scholars have reported that expression of TJ proteins such as ZO-1 and occludin is decreased markedly in UC patients and mice with DSS-induced colitis (Poritz et al., 2011; Zhang et al., 2019). Hence, regulation of expression of TJ proteins may be important in UC treatment. After treatment with SCE, the contents of ZO-1 and occludin in DSS-induced UC mice increased significantly, which suggests that SCE exerts an anti-UC effect by improving intestinal barrier function.

It has been pointed out that dysfunction of the intestinal microbiota is associated with UC pathogenesis (Costello et al., 2019). Combined with the study of intestinal bacteria, the relative abundance of beneficial gut bacteria decreased, while that of pathogenic gut bacteria increased in DSS-induced UC mice, in comparison with those in healthy mice. In contrast, SCE promoted recovery of the abundance of beneficial bacteria and harmful bacteria, as evidenced by a higher abundance of bacteria from the phylum Bacteroidetes and genera *Turicibacter*, *Ruminococcus*, *Odoribacter*, *Oscillospira* and *Parabacteroides*, and lower abundance of

bacteria from the phylum Actinobacteria and genera *Streptococcus* and *Adlercreutzia*. It has been reported that intestinal bacteria of the phylum Bacteroidetes have a protective effect against UC; they synthesize conjugated linoleic acid and contribute to the homeostasis of intestinal flora (Nomura et al., 2021). The results of this study show that the relative abundance of the *Bacteroidetes* enhanced prominently in the SCE administration treatment group compared with that in the DSS-induced UC group. The significantly decreased relative abundance of bacteria of the phylum Actinobacteria after SCE intervention was consistent with data from other studies (Liu et al., 2021; Niu et al., 2021). Bacteria of the genus *Odoribacter* can produce butyrate, which is beneficial for gut health (Cheng et al., 2018). The number of bacteria of the genus *Oscillospira* is significantly lower in UC patients than that in healthy people, which is positively correlated with intestinal health (Xu et al., 2021). *Streptococcus* species are harmful bacteria that cause dental caries. The number of dental caries in patients with inflammatory bowel disease increased in contrast with healthy people (Santor et al., 2017). Above evidences suggest that SCE may improve DSS-induced UC by regulating the composition of the intestinal microbiota.

By comparing the chemical composition of our extract with that of other products (extracts) previously studied, we found that lignans are the main components of *Schisandra chinensis* extracts. These lignans, such as schisandrin, schizandrol B, schisantherin A, deoxyschisandrin, schisandrin B are commonly detected components of *Schisandra chinensis* extracts (Kosman et al., 2014). Previous pharmacokinetic studies have shown that the prototypes of six lignans such as schizandrol A, schizandrol B, schisantherin A, deoxyschisandrin, schisandrin B, and schisandrin C can even be detected in blood and liver more than 6 h after administration of *Schisandra chinensis* lignans, indicating that the elimination phase of these six lignans were slower and the oral exposure time were relatively longer (Wang et al., 2020; Yang et al., 2022). These lignans may be the potential effective substances to exert efficacy. Lignan-based compounds presented in SCE, such as deoxyschisandrin and schisandrin B, have been shown to exert anti-colitis effects. Deoxyschisandrin could alleviate the symptoms of UC in mice through

inhibiting inflammatory response and apoptosis (Zhang et al., 2016). Schisandrin B could exert beneficial effects on anti-colitis through suppressing the secretion of proinflammatory cytokines and regulating of AMPK/Nrf2/NLRP3 inflammasome pathway (Zhang et al., 2021). We also indicated that the effects of SCE in treating UC might be through inhibition of the TLR4/NF- κ B/NLRP3 inflammasome signaling pathway and modulation of the intestinal microbiota.

Taken together, we found that SCE may be a potential multi-target drug candidate for the treatment of UC. On the one hand, its therapeutic effects are associated with inhibition of inflammation-related pathways to relieve inflammatory response. On the other hand, SCE could regulate the composition of GM to maintain the gut health. Dysfunction of the GM is associated with UC pathogenesis. Therefore, this study demonstrated that SCE may have a health benefit for the clinical treatment of human UC involving GM.

5. Conclusions

We demonstrated that SCE can ameliorate DSS-induced colitis, and that its protective effects may be associated with: (i) inhibition of secretion of proinflammatory cytokines; (ii) improvement of oxidative stress; (iii) suppression of the TLR4/NF- κ B/NLRP3 inflammasome signaling pathway; (iv) protection of intestinal-barrier integrity; (v) GM modulation. Dibenzocyclooctene lignans, the main phytochemicals in SCE could be developed into promising candidates for UC treatment and merit further research attention.

Ethics statement

The animal experiments described herein were approved by the Ethics Committee of Wuxi TCM Hospital Affiliated to Nanjing University of Chinese Medicine.

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CRediT authorship contribution statement

Zhenhua Bian: Carried out the experiments, Analyzed the data, Wrote the manuscript. **Yuwen Qin:** Carried out the experiments, Analyzed the data, Revised the manuscript. **Lin Li:** Carried out the experiments, Analyzed the data, Revised the manuscript. **Lianlin Su:** Carried out the experiments, Analyzed the data. **Chenghao Fei:** Carried out the experiments, Analyzed the data. **Yu Li:** Carried out the experiments, Analyzed the data. **Minmin Hu:** Carried out the experiments, Analyzed the data. **Xiaowei Chen:** Carried out the experiments, Analyzed the data. **Wei Zhang:** Carried out the experiments, Analyzed the data. **Chunqin Mao:** Carried out the experiments, Analyzed the data. **Xiaohang Yuan:** Designed and directed the research. **Tulin Lu:** Designed and directed the research. **De Ji:** Designed and directed the research.

Declaration of competing interest

None of the authors have any conflicts of interest to declare.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2022.115570>.

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